Dalle Molle Institute for Artificial Intelligence
USI – SUPSI

Learning to Identify Dynamical Systems with Meta Learning

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SUPSI

Motivation

- Machine learning models can effectively describe a wide variety of dynamical systems
- However, standard training is computationally expensive and data-hungry
- Can we automate the design of algorithms, optimizers, model structures for SYSID problems?
- This is meta learning, aka learning to learn!

The “sines” toy example

- Multiple datasets with structure:

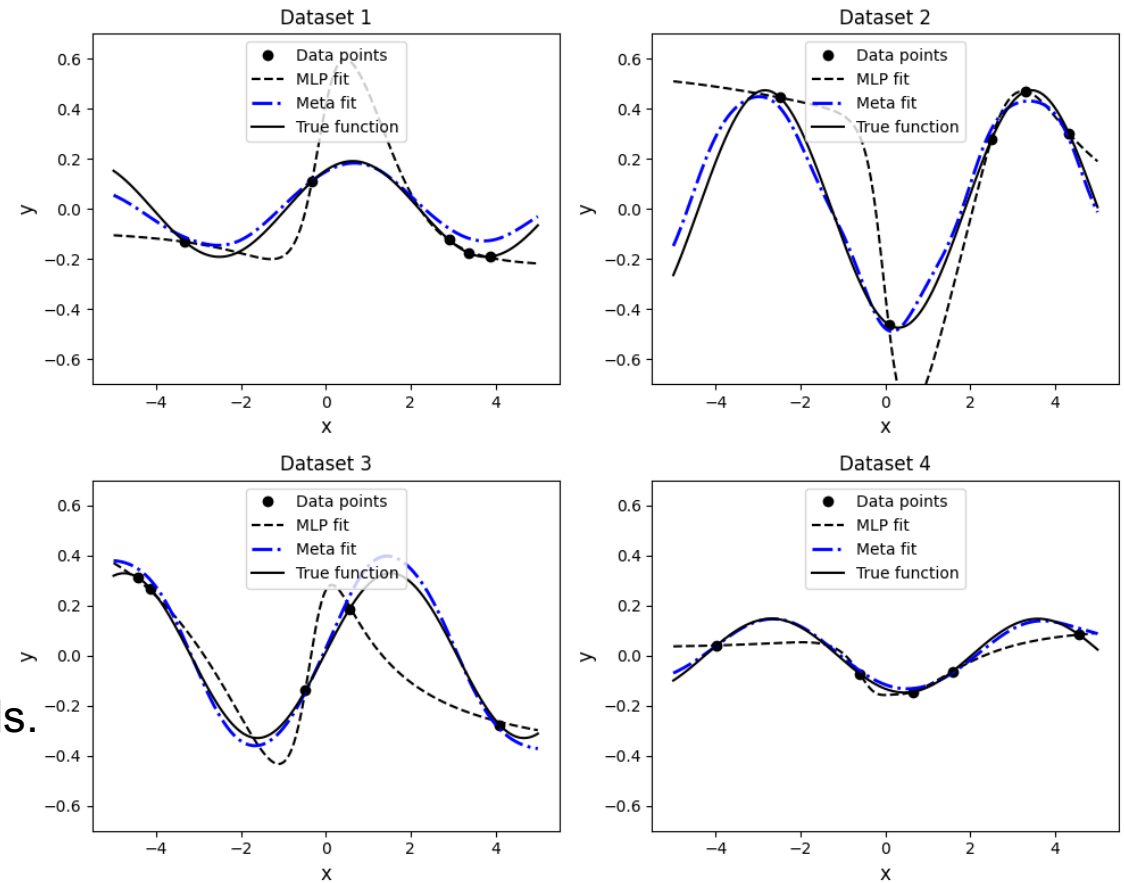
$$y(x) = A\sin(x + \alpha)$$

$K = 5$ points each, random A and α.

- Training MLPs separately on each dataset, we overfit the $K = 5$ points.

- What if we see $K = 5$ points across many datasets? Can we learn the pattern?

- Results with existing meta learning methods.



T. Hospedales et al. Meta-Learning in Neural Networks: A survey.
IEEE Transactions on Patten Analysis and Machine Intelligence, 2022.

https://github.com/forgi86/pytorch-meta-learning

https://github.com/forgi86/jax-tutorial

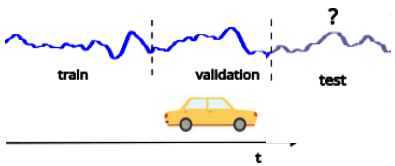
From standard to meta system identification

- System Identification
  - **Dataset**: I/O trajectory $D = (u_{1:N}, y_{1:N})$ of one particular system
  - **Objective**: fit a model that generalizes to new inputs

$$\theta = \mathrm{Alg}(D) \qquad \min_{\theta} \|y_{1:N} - M_\theta(u_{1:N})\|^2$$

$$\hat{y}^*_{1:N} = M_\theta(u^*_{1:N})$$

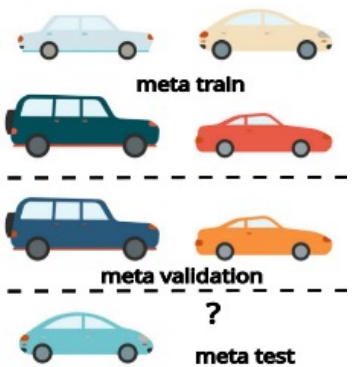
  - Alg *manually* designed and tuned



- Meta System Identification
  - **Meta dataset**: many datasets $\mathfrak{D} = \{D^{(1)}, D^{(2)}, \ldots\}$ from similar systems
  - **Objective**: learn an algorithm that generalizes to new systems

$$\theta = \mathrm{Alg}_\phi(D) \qquad \min_{\phi} \sum_i \mathcal{J}(D^{(i)}, \mathrm{Alg}_\phi)$$

$$\hat{y}^*_{1:N} = M_\theta(u^*_{1:N})$$



  - What does it mean to learn an algorithm? What is the search space?

From algorithm learning to in-context learning

- **Meta dataset**: collection of datasets $\mathfrak{D} = \{D^{(1)}, D^{(2)}, \ldots\}$ from different yet similar systems
- **Objective**: learn an algorithm $\Rightarrow \min_{\phi} \sum_i \mathcal{J}(D^{(i)}, \mathrm{Alg}_\phi)$

algorithm learning

$$\theta = \mathrm{Alg}_\phi(D)$$

$$\hat{y}^* = M_\theta(u^*)$$

in-context learning

$$\hat{y}^* = \mathcal{M}_\phi(u^*, D)$$

$\mathcal{M}_\phi$ meta model

$$\mathrm{Alg}_\phi(D) = \arg\min_\theta \|y - M_\theta(u)\|^2 + \phi \|\theta\|^2$$

$$\mathrm{Alg}_\phi(D) = \arg\min_\theta \|y - M_\theta(u)\|^2 + \lambda \|\theta - \phi\|^2$$

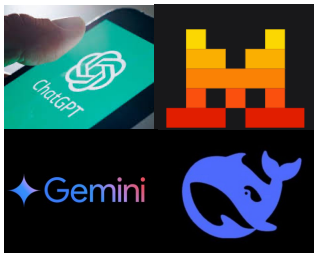
$$\mathrm{Alg}_\phi(D) = \phi - \gamma \nabla_\phi \|y - M_\phi(u)\|^2 \quad \text{(MAML)}$$

$$\mathrm{Alg}_\phi(D) = \mathrm{BlackBox}_\phi(D)$$

In-context learning in Large Language Models

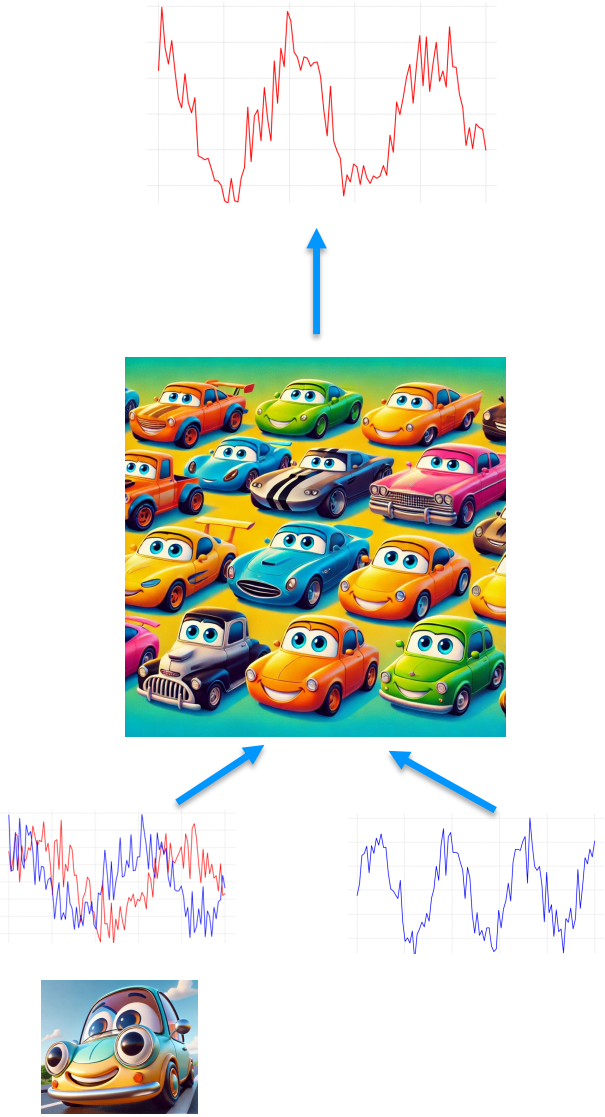
Answer from LeChat

- **Vieux Lyon**: *Vieux Lyon is one of Europe's largest Renaissance districts and a UNESCO World Heritage site. Wander through its narrow, cobbled streets (traboules) and admire the stunning architecture…*
- *…*
- **Basilique Notre-Dame de Fourvière**: *Perched atop Fourvière Hill, this 19th-century basilica is a symbol of Lyon. Its breathtaking Byzantine and Gothic Revival architecture is matched only by the panoramic views of the city below…*



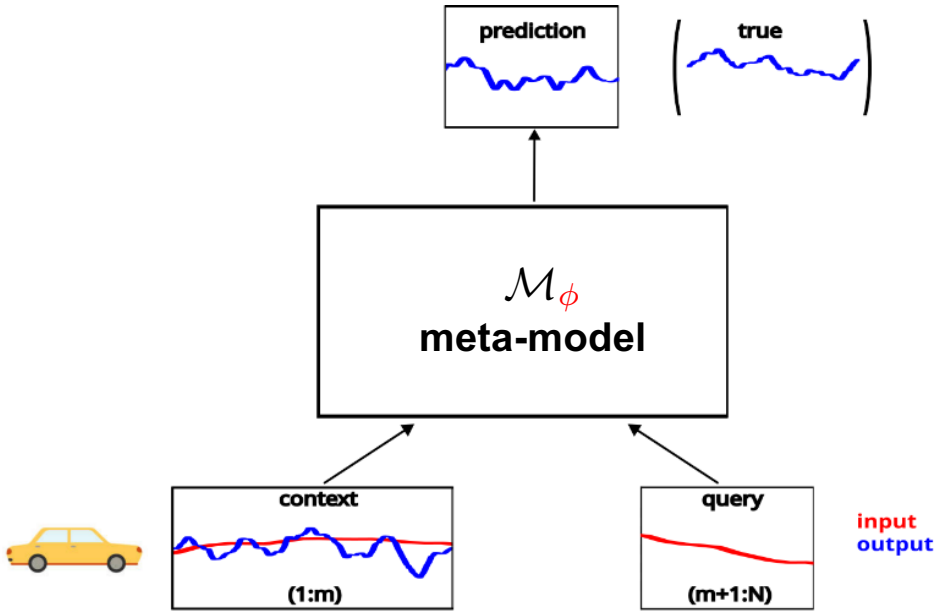
Context
I'm passionate about history and planning a trip to Lyon

Query
Can you recommend some places to visit?



In-context system identification

- Input/output context data, input query → predicted query output:



$$\hat{y}_{m+1:N} = \mathcal{M}_\phi(u_{m+1:N}, D)$$

$D = (u_{1:m}, y_{1:m})$ is the context

$u_{m+1:N}$ is the query

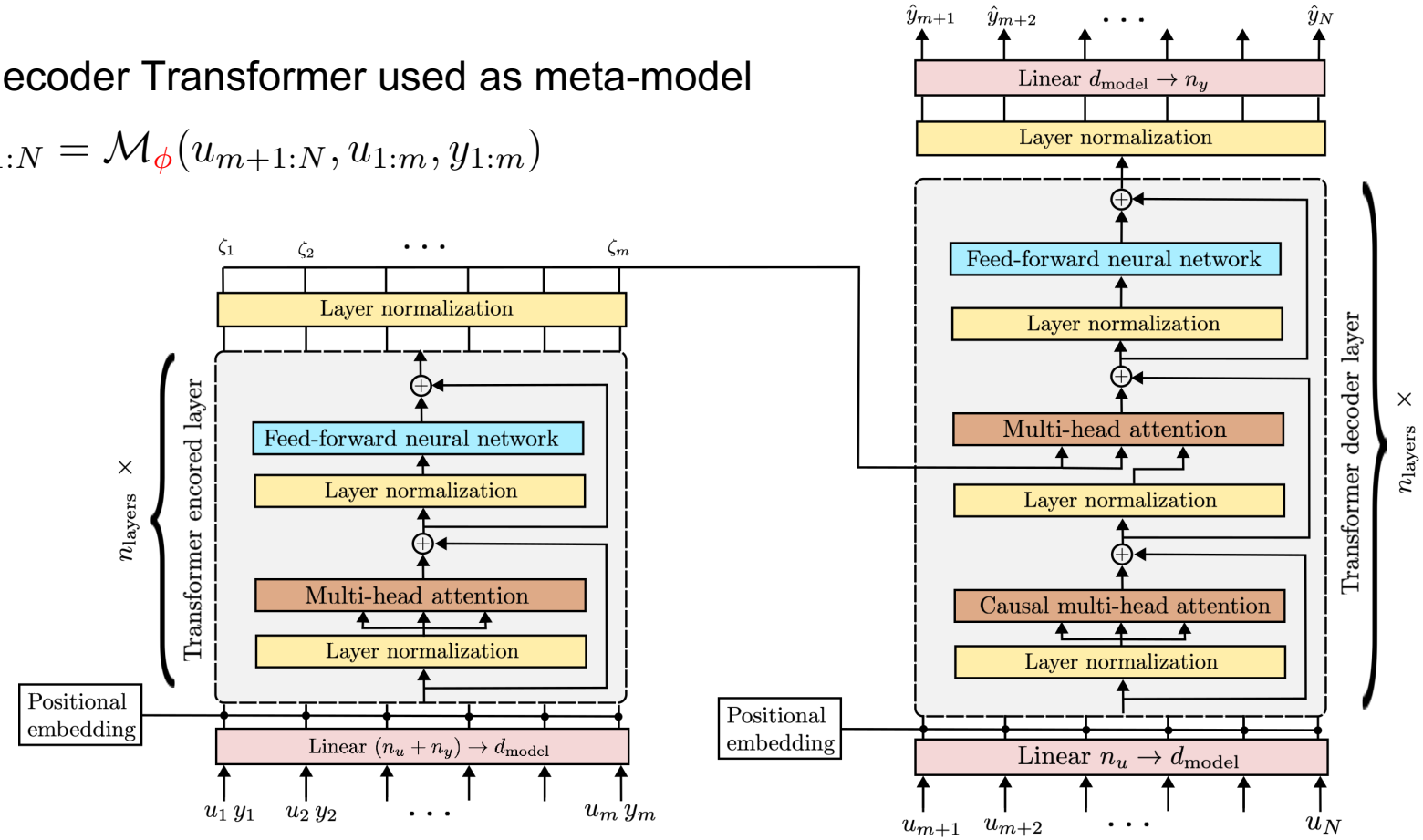
$\hat{y}_{m+1:N}$ is the answer

M. Forgione, F. Pura & D. Piga. From system models to class models: An in-context learning paradigm. *IEEE L-CSS, 2023.*

Meta model architecture

Encoder-decoder Transformer used as meta-model

$$\hat{y}_{m+1:N} = \mathcal{M}_{\phi}(u_{m+1:N}, u_{1:m}, y_{1:m})$$



Extended to handle longer sequences, provide uncertainty bounds, handle initial conditions.

M. Rufolo, D. Piga, G. Maroni & M. Forgione. Enhanced Transformer architecture for in-context learning of dynamical systems. ECC 2025.

Meta model training

STEPS

1. Collect a meta dataset of similar systems (from a simulator)

$$\mathcal{D} = \{D^{(1)}, D^{(2)}, \dots\}$$

2. Train the meta-model off-line:

$$\hat{\phi} = \arg\min_{\phi} \sum_{i} \left\| y^{(i)}_{m+1:N} - \mathcal{M}_{\phi}\left(u^{(i)}_{m+1:N},\ u^{(i)}_{1:m}, y^{(i)}_{1:m}\right) \right\|^2$$

After training, predict on a new dataset D^{new} in a zero-shot fashion:

$$\hat{y}^{*}_{m+1:N} = \mathcal{M}_{\hat{\phi}}(u^{*}_{m+1:N}, u^{\text{new}}_{1:m}, y^{\text{new}}_{1:m})$$

- Technically, just a supervised learning problem – solved by gradient descent or similar
- The trained $\mathcal{M}_{\hat{\phi}}$ is a *specialized* question answering system – a kind of SYSID GPT.

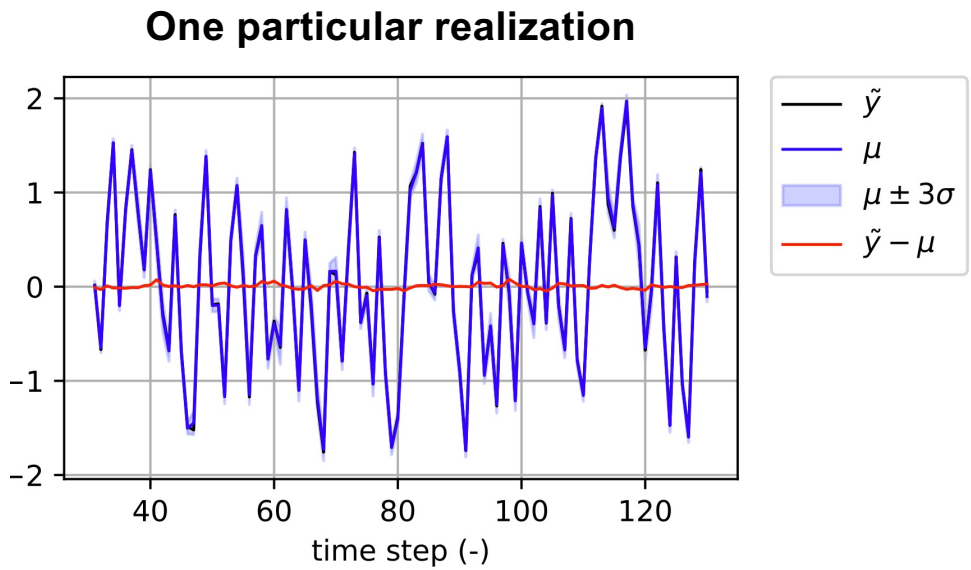
Meta model training cont'd

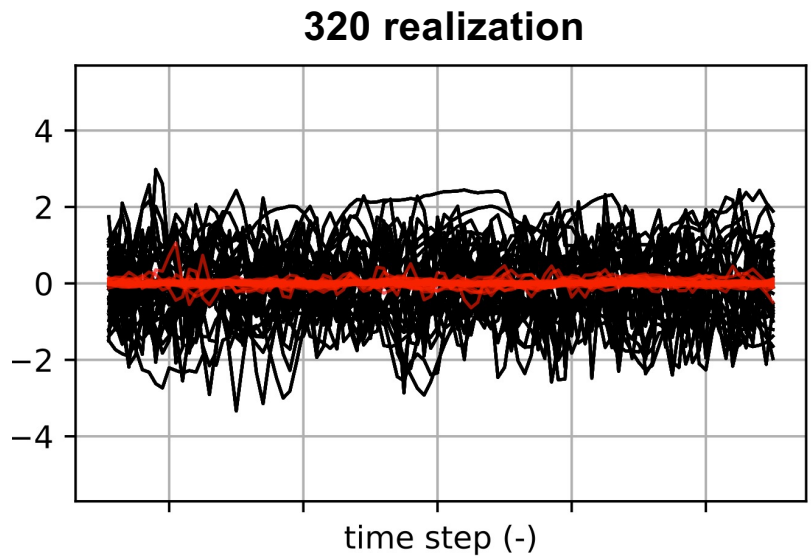
Wiener-Hammerstein Systems

- Sequential LTI → $F(\cdot)$ → LTI
- Random LTI of order ≤ 5
- $F(\cdot)$: random feed-forward NN

- Meta training on ~100 M simulated Wiener-Hammerstein systems
- The input is a multisine signal with random phase and spectrum
- White noise added to the simulated output
- Transformer with ~5 M parameters, training time ~1 day on an Nvidia 4090 GPU…

Testing – Synthetic datasets in the WH class

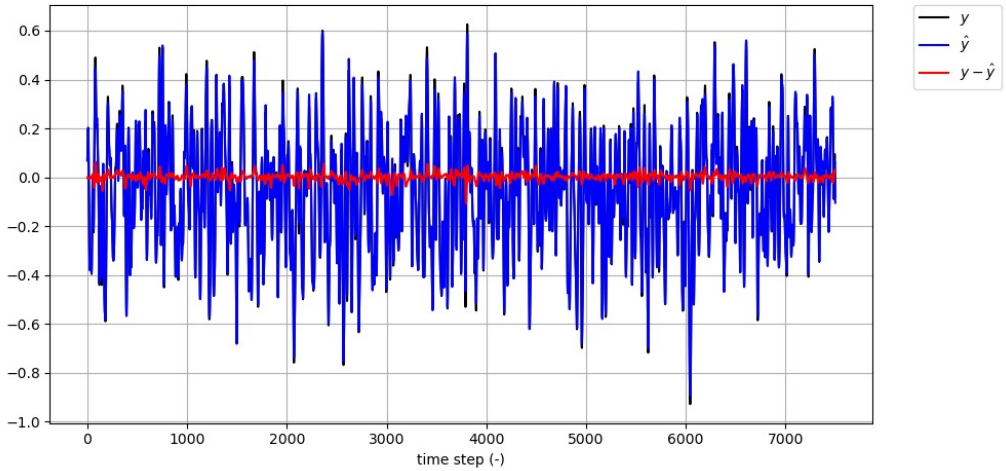


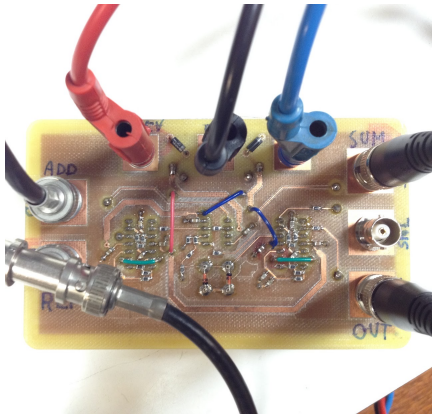


Average normalized RMSE ~ 0.05 on *new data* (from the same synthetic distribution)

Testing – Nonlinear benchmarks (www.nonlinearbenchmark.org)

- Real Wiener-Hammerstein benchmark



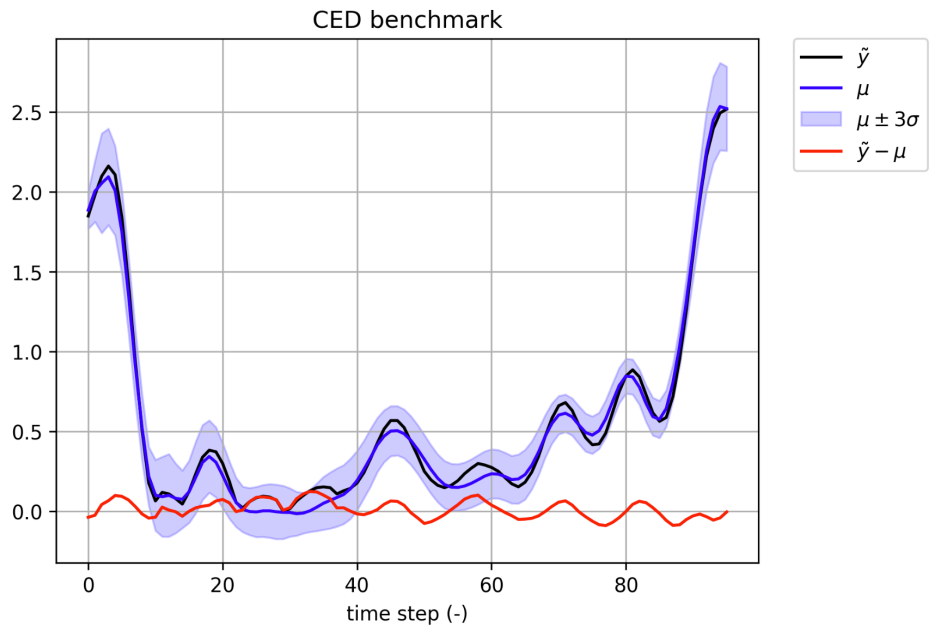


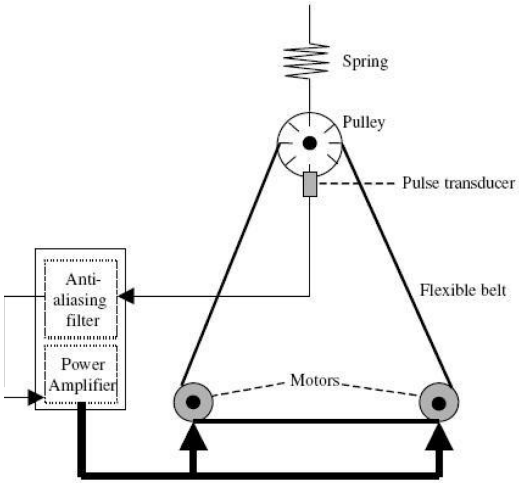
RMSE * 1000	Inference time (s)
1.4	5.91

Comparable with SOTA, but in zero-shot (no training)!

Testing – Nonlinear benchmarks

- Coupled Electric Drives (CED)



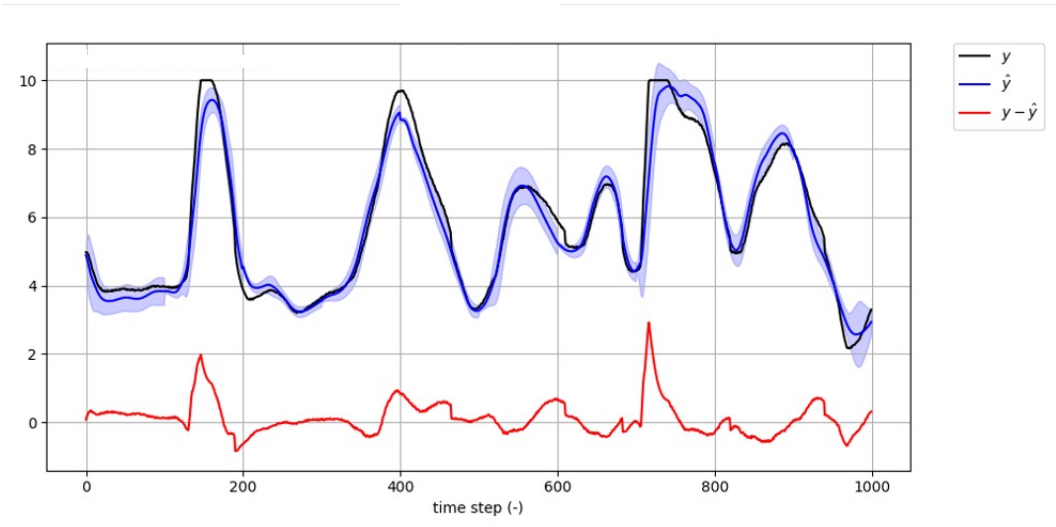


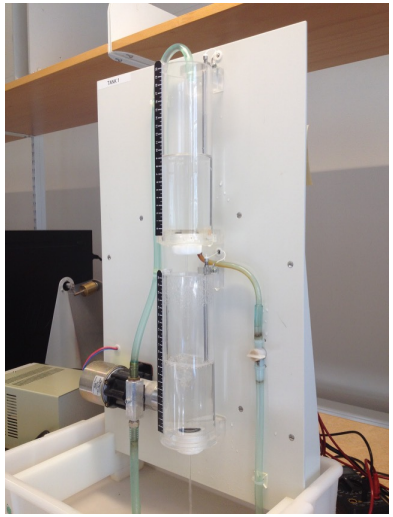
RMSE	Inference time (s)
[0.068,0.055]	0.17

Comparable with SOTA, but in zero-shot (no training)!

Testing – Nonlinear benchmarks

- Cascaded tanks with Overflow



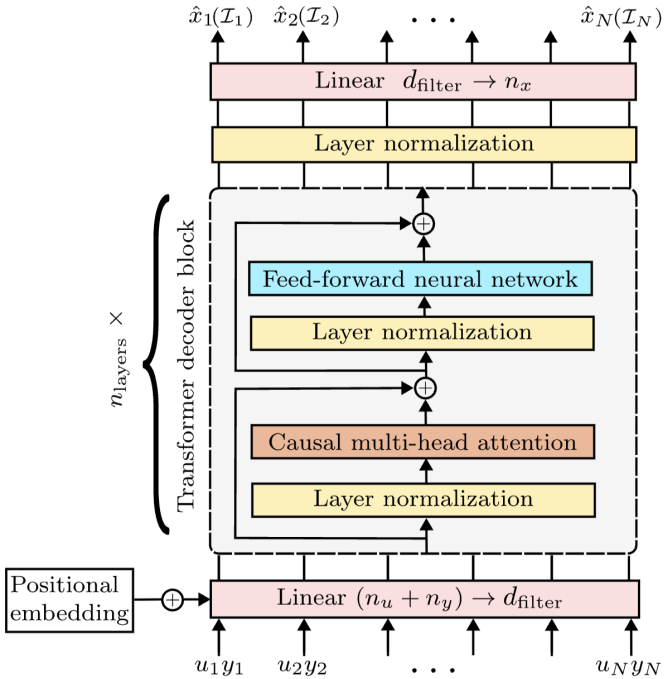


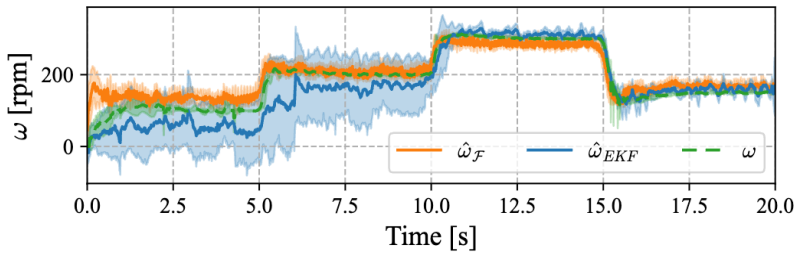
RMSE	Inference time (s)
0.429	0.22

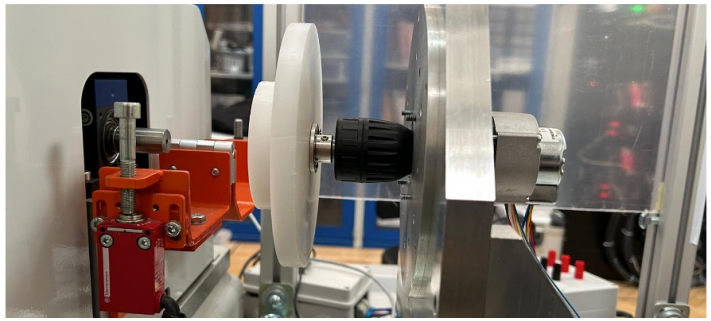
Comparable with SOTA, but in zero-shot (no training)!

Other objectives: meta state estimation

- A meta state estimator that generalizes to different systems from the context







- Experiments on speed estimation of BLDC motors

R. Busetto, V. Breschi, M. Forgione, D. Piga & S. Formentin. In-Context Learning of State Estimators. SYSID 2024.

A. Colombo, R. Busetto, V. Breschi, M. Forgione, D. Piga. & S. Formentin. In-Context Learn. for Zero-Shot Speed Est. of BLDC Motors. arXiv 2025.

Whitening the meta-learning box

Algorithm learning

$$\hat{y}^* = M_{\mathrm{Alg}_\phi(D)}(u^*)$$

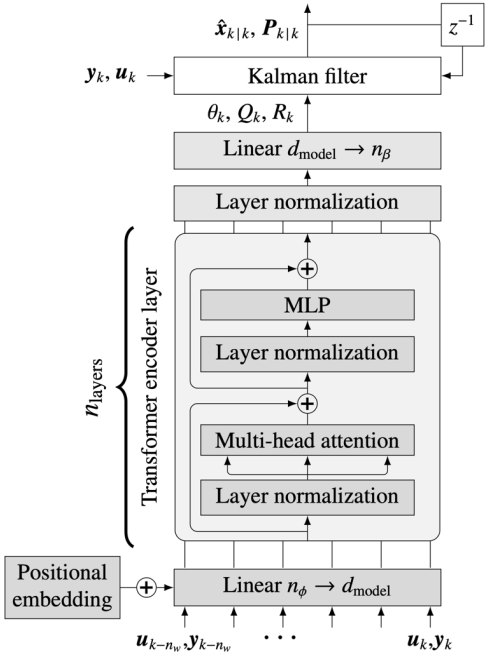
In-context learning

$$\hat{y}^* = \mathcal{M}_\phi(u^*, D)$$

- In-context learning: prediction is a black-box function of dataset D and new input u^*.
- Other extreme: learn hyper-parameters ϕ of algorithm Alg_ϕ that returns model parameters
  - Learn a regularizer, initialization of gradient descent (MAML), learning rate, etc
- What's in-between?

Meta state estimation: whitening the box

- Transformer returns covariances + physical parameters for a KF instead of predictions directly.



Collaboration with H. Koide & G. Mercère. “Transformer-aided Kalman filter for in-context recursive state estimation”, in preparation.

Conclusions

- Meta learning for system identification (estimation, optimization, …)
- In-context learning:
  - Off-line: learn to map datasets into predictions
  - On-line: I/O context, query input → query output (zero shot)
- Whitening the meta-learning box

Active/future research

- Theoretical guarantees
- Larger meta models for wider classes of systems…
  or smaller ones tailored for specific applications?
- Algorithm discovery with language models

THANKS FOR YOUR ATTENTION!